

EXAMEN DAW FEBRERO

Endika Peña Alonso

Apartado A

Ejercicio 1: Instalación de apache2 y tomcat

Para que la configuración sea más sencilla instalaré openssh-server, de forma que pueda conectarme desde el equipo host.

sudo apt install openssh-server

```
After this operation, 6,853 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sftp-server amd64 1:9.6p1-3ubuntu13.14 [37.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libwrap0 amd64 7.6.q-33 [47.9 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-server amd64 1:9.6p1-3ubuntu13.14 [510 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 ncurses-term all 6.4+20240113-1ubuntu2 [275 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 ssh-import-id all 5.11-0ubuntu2.24.04.1 [10.1 kB]
Fetched 880 kB in 1s (813 kB/s)
Preconfiguring packages ...
Selecting previously unselected package openssh-sftp-server.
(Reading database ... 84573 files and directories currently installed.)
Preparing to unpack .../openssh-sftp-server_1:9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-sftp-server (1:9.6p1-3ubuntu13.14) ...
Selecting previously unselected package libwrap0:amd64.
Preparing to unpack .../libwrap0_7.6.q-33_amd64.deb ...
Unpacking libwrap0:amd64 (7.6.q-33) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1:9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu13.14) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.4+20240113-1ubuntu2_all.deb ...
Unpacking ncurses-term (6.4+20240113-1ubuntu2) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu2.24.04.1_all.deb ...
Unpacking ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.14) ...
Setting up ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up libwrap0:amd64 (7.6.q-33) ...
Setting up ncurses-term (6.4+20240113-1ubuntu2) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...

Creating config file /etc/ssh/sshd_config with new version
Created symlink /etc/systemd/system/sockets.target.wants/ssh.socket → /usr/lib/systemd/system/ssh.socket.
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket → /usr/lib/systemd/system/ssh.socket.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

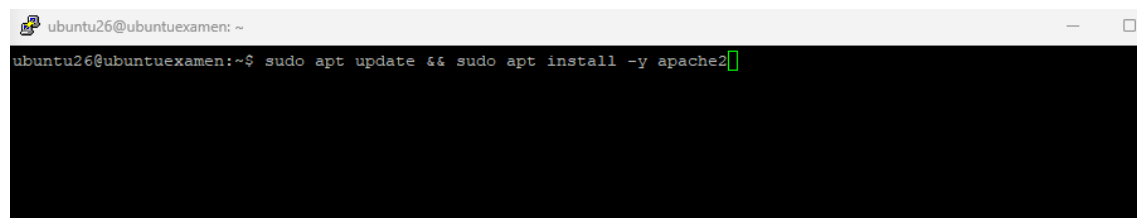
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

Actualizar repositorios e instalación de apache2.

sudo apt update && sudo apt install -y apache2



```
ubuntu26@ubuntuetexamen: ~
ubuntu26@ubuntuetexamen:~$ sudo apt update && sudo apt install -y apache2
```

Instalación de JDK y servidor de aplicaciones tomcat.

sudo apt install -y default-jdk curl

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu26@ubuntuexamen:~$ sudo apt install -y default-jdk curl
```

Creación del usuario y grupo para tomcat.

sudo groupadd tomcat && useradd -s /bin/false -g tomcat -d /opt/tomcat tomcat

```
ubuntu26@ubuntuexamen: ~
ubuntu26@ubuntuexamen:~$ sudo useradd -s /bin/false -g tomcat -d /opt/tomcat tomcat
```

Descargamos la versión 10 de tomcat desde

<https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.52/bin/apache-tomcat-10.1.52.tar.gz>

```
ubuntu26@ubuntuexamen: /tmp
ubuntu26@ubuntuexamen:/tmp$ curl -O https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.52/bin/apache-tomcat-10.1.52.tar.gz
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 13.6M  100 13.6M    0     0  9655k      0  0:00:01  0:00:01 --:--:-- 9654k
ubuntu26@ubuntuexamen:/tmp$
```

Creamos la carpeta de tomcat y le cambiamos el propietario

sudo mkdir -p /opt/tomcat && sudo chown tomcat:tomcat -R /opt/tomcat

```
ubuntu26@ubuntuexamen:/tmp$ sudo mkdir -p /opt/tomcat && sudo chown tomcat:tomcat -R /opt/tomcat
ubuntu26@ubuntuexamen:/tmp$
```

Desempaquetamos el código de tomcat y volvemos a asignar el propietario y grupo.

sudo tar -xzf apache-tomcat-10.1.52.tar.gz -C /opt/tomcat --strip-components=1

sudo chown tomcat:tomcat -R /opt/tomcat

```
ubuntu26@ubuntuexamen: /tmp
ubuntu26@ubuntuexamen:/tmp$ sudo chown tomcat:tomcat -R /opt/tomcat
ubuntu26@ubuntuexamen:/tmp$
```

Definiendo el servicio de tomcat para que hable con el sistema.

`sudo nano /etc/systemd/system/tomcat.service`

```
ubuntu26@ubuntuexamen: /tmp
ubuntu26@ubuntuexamen:/tmp$ sudo chown tomcat:tomcat -R /opt/tomcat
ubuntu26@ubuntuexamen:/tmp$ sudo nano /etc/systemd/system/tomcat.service
ubuntu26@ubuntuexamen:/tmp$ sudo nano /etc/systemd/system/tomcat.service
ubuntu26@ubuntuexamen:/tmp$ cat /etc/systemd/system/tomcat.service
[Unit]
Description=Apache Tomcat Web Application Container
After=network.target

[Service]
Type=forking

Environment="JAVA_HOME=/usr/lib/jvm/java-1.21.0-openjdk-amd64"
Environment="CATALINA_PID=/opt/tomcat/temp/tomcat.pid"
Environment="CATALINA_HOME=/opt/tomcat/"
Environment="CATALINA_BASE=/opt/tomcat/"
Environment="CATALINA_OPTS=-Xms512M -Xmx1024M -server -XX:+UseParallelGC"
Environment="JAVA_OPTS=-Djava.awt.headless=true -Djava.security.egd=file:/dev/.urandom"

ExecStart=/opt/tomcat/bin/startup.sh
ExecStop=/opt/tomcat/bin/shutdown.sh

User=tomcat
Group=tomcat
UMask=0007
RestartSec=10
Restart=always

[Install]
WantedBy=multi-user.target
```

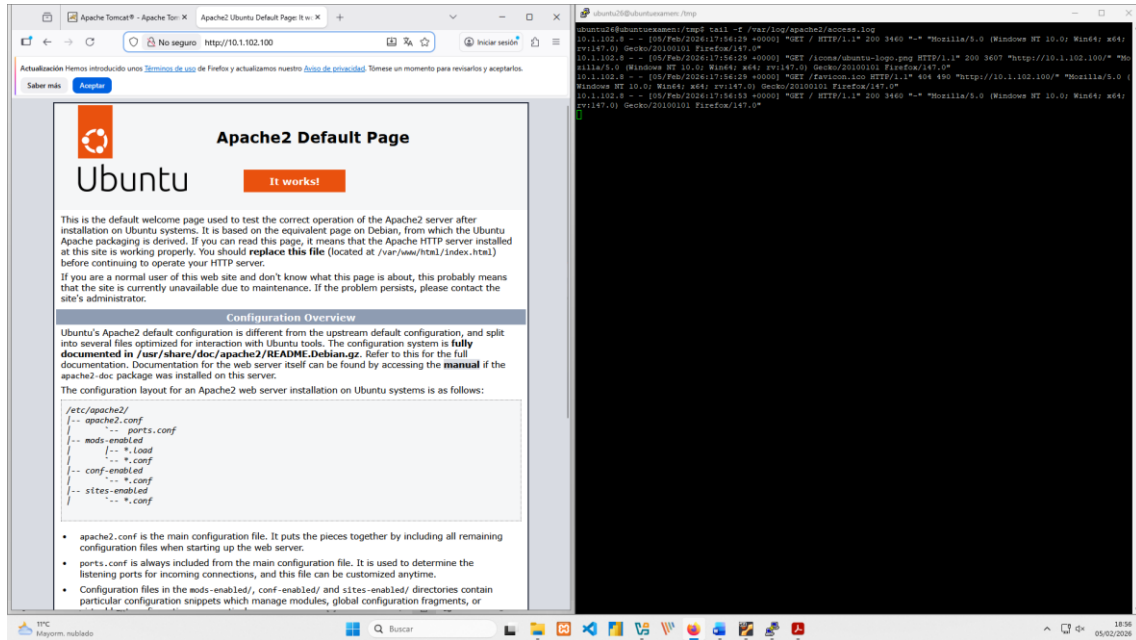
Recargamos el Daemon para que el sistema lo encuentre y activamos el servicio y lo habilitamos para que arranque solo entre reinicios.

```
ubuntu26@ubuntuexamen: /tmp
ubuntu26@ubuntuexamen:/tmp$ sudo systemctl daemon-reload
ubuntu26@ubuntuexamen:/tmp$ sudo systemctl start tomcat
ubuntu26@ubuntuexamen:/tmp$ sudo systemctl status tomcat
● tomcat.service - Apache Tomcat Web Application Container
   Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: enabled)
   Active: active (running) since Thu 2026-02-05 17:54:41 UTC; 8s ago
     Process: 4829 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=0/SUCCESS)
    Main PID: 4836 (java)
      Tasks: 32 (limit: 4602)
   Memory: 142.5M (peak: 143.0M)
      CPU: 12.148s
   CGroup: /system.slice/tomcat.service
           └─4836 /usr/lib/jvm/java-1.21.0-openjdk-amd64/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/

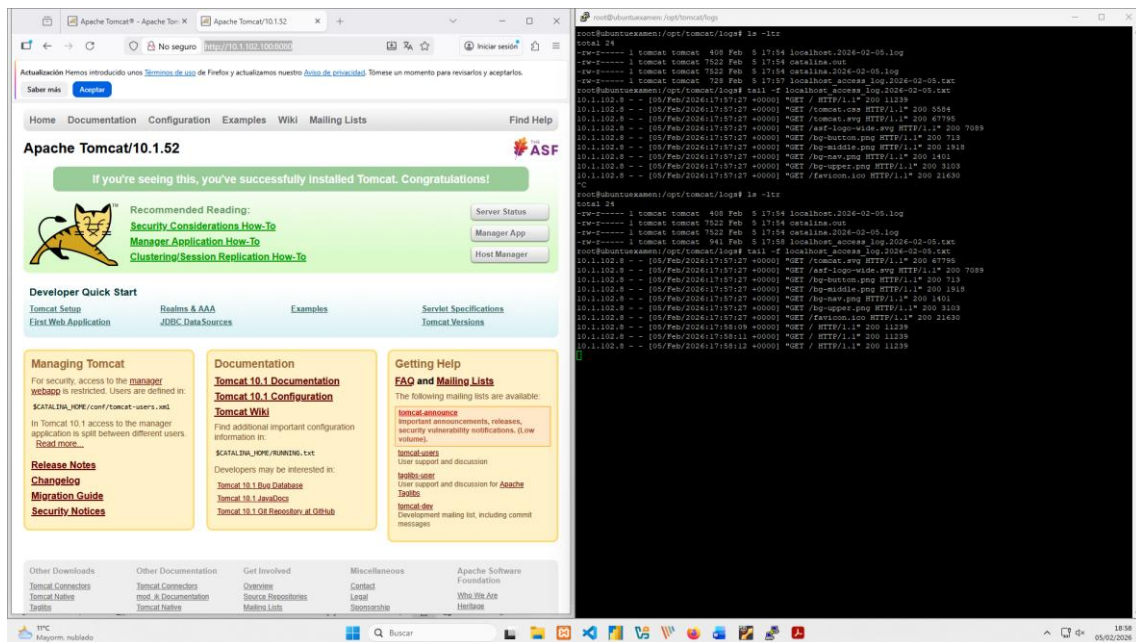
Feb 05 17:54:41 ubuntuexamen systemd[1]: Starting tomcat.service - Apache Tomcat Web Application Container...
Feb 05 17:54:41 ubuntuexamen startup.sh[4829]: Tomcat started.
Feb 05 17:54:41 ubuntuexamen systemd[1]: Started tomcat.service - Apache Tomcat Web Application Container.
ubuntu26@ubuntuexamen:/tmp$ sudo systemctl enable tomcat
Created symlink /etc/systemd/system/multi-user.target.wants/tomcat.service → /etc/systemd/system/tomcat.service.
ubuntu26@ubuntuexamen:/tmp$
```

Ejercicio 2: Comprobaciones

Comprobación de funcionamiento de apache2



Comprobación de funcionamiento de tomcat



Apartado B

Ejercicio 1: Estructura y Virtualhost

Creación de carpetas y fichero.

```
sudo mkdir -p /var/www/startup-tech/publico
```

```
sudo nano /var/www/startup-tech/publico/index.html
```

```
ubuntu26@ubuntuexamen:/tmp$ tree /var/www
/var/www
├── html
│   └── index.html
└── startup-tech
    ├── publico
    │   └── index.html
    └── 
```

Configuración de VHOST

Voy a copiar el 000-default.conf y lo voy a usar para modificarlo para mi sitio.

```
sudo cp 000-default.conf startup-tech.local.conf
```

```
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ sudo nano startup-tech.local.conf
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ sudo systemctl reload apache2
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ sudo nano startup-tech.local.conf
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ cat startup-tech.local.conf
<VirtualHost *:80>
    ServerName startup-tech.local
    ServerAlias www.startup-tech.local
    # The ServerName directive sets the request scheme, hostname and port that
    # the server uses to identify itself. This is used when creating
    # redirection URLs. In the context of virtual hosts, the ServerName
    # specifies what hostname must appear in the request's Host: header to
    # match this virtual host. For the default virtual host (this file) this
    # value is not decisive as it is used as a last resort host regardless.
    # However, you must set it for any further virtual host explicitly.
    #ServerName www.example.com

    ServerAdmin endika@localhost
    DocumentRoot /var/www/startup-tech/

    # Available loglevels: trace8, ..., tracel, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

    ErrorLog ${APACHE_LOG_DIR}/startup_tech_local-error.log
    CustomLog ${APACHE_LOG_DIR}/startup_tech_local-access.log combined

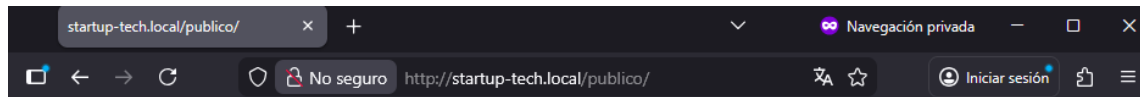
    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```

Habilitamos el nuevo sitio web.

`sudo a2ensite startup-tech.local`

```
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ sudo a2ensite startup-tech.local
Enabling site startup-tech.local.
To activate the new configuration, you need to run:
    systemctl reload apache2
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$
```

Prueba de funcionamiento después de modificar el /etc/host de Windows incluyendo el dominio tanto con www como sin el.



Bienvenido a Startup Tech - ENDIKA

Ejercicio 2 Seguridad SSL y Redirección de http a https

Primero activamos el modulo de rewrite que nos permite realizar las redirecciones de http a https

```
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ sudo a2enmod rewrite
Enabling module rewrite.
To activate the new configuration, you need to run:
    systemctl restart apache2
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$
```

Vamos modificar el mismo archivo del vhost de startup-tech.local.conf para incluir en el 2 virtual host uno para SSL y otro sin.

```
ubuntu26@ubuntuexamen: /etc/apache2/sites-available
GNU nano 7.2 startup-tech.local.conf *
<VirtualHost *:80>
    ServerName startup-tech.local
    ServerAlias www.startup-tech.local

    ServerAdmin endika@localhost

    CustomLog ${APACHE_LOG_DIR}/startup_tech_local-access.log combined

    RewriteEngine On
    #Redireccion de trafico http a https
    RewriteCond %{HTTPS} off
    RewriteRule ^/(.*)$ https://%{SERVER_NAME}/$1 [R=301,L]
</VirtualHost>

<VirtualHost *:443>
    ServerName startup-tech.local
    ServerAlias www.startup-tech.local

    ServerAdmin endika@localhost

    DocumentRoot /var/www/startup-tech/publico

    SSLEngine on
    SSLCertificateFile /etc/apache2/ssl_certs/startup-tech.local.crt
    SSLCertificateKeyFile /etc/apache2/ssl_certs/startup-tech.local.key

    DirectoryIndex index.html

    ErrorLog ${APACHE_LOG_DIR}/startup_tech_local-error.log
    CustomLog ${APACHE_LOG_DIR}/startup_tech_local-access.log combined
</VirtualHost>
```

Antes de reiniciar el servicio de apache vamos a revisar si tenemos el modulo de SSL activo y vamos a generar el certificado en las rutas que hemos definido.

```
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ sudo a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
    systemctl restart apache2
```

Generamos el certificado autofirmado

```
openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout
/etc/apache2/ssl_certs/startup-tech.local.key -out /etc/apache2/ssl_certs/startup-
tech.local.crt -subj
"/C=ES/ST=LaRioja/L=Villamediana/O=FPDRioja/OU=DAW_Examen/CN=startup-
tech.local" -addext "subjectAltName = DNS:startup-tech.local,DNS:www.startup-
tech.local"
```

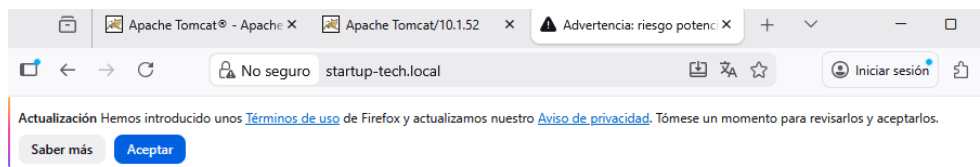
Se han creado los ficheros para almacenar la clave publica y privada y hemos modificado tanto el propietario para que sea el usuario www-data y los permisos para que sean restrictivos 400 para el .key y 440 para .crt

```
ubuntu26@ubuntueexamen:/etc/apache2/sites-available$ sudo chmod 400 /etc/apache2/ssl_certs/startup-tech.local.key
ubuntu26@ubuntueexamen:/etc/apache2/sites-available$ sudo chmod 440 /etc/apache2/ssl_certs/startup-tech.local.crt
ubuntu26@ubuntueexamen:/etc/apache2/sites-available$ tree /etc/apache2/ssl_certs
/etc/apache2/ssl_certs
├── startup-tech.local.crt
└── startup-tech.local.key
1 directory, 2 files
```

Ahora reiniciamos con el modulo SSL activo y hacemos la comprobación

```
sudo systemctl restart apache2
```

El navegador ya nos advierte que el certificado no es “confiable”



Advertencia: riesgo potencial de seguridad a continuación

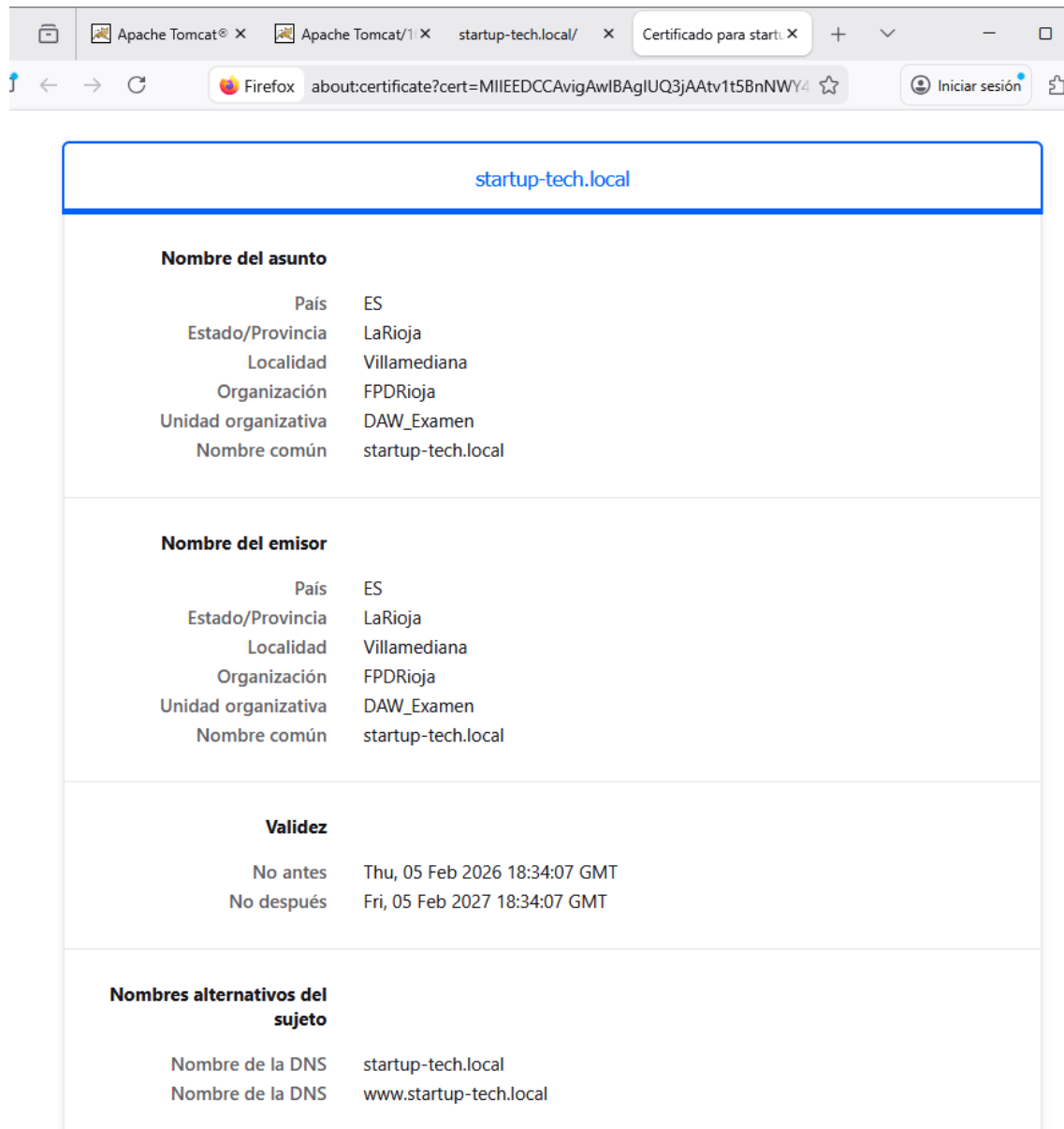
Firefox ha detectado una posible amenaza de seguridad y no ha cargado **startup-tech.local**. Si visita este sitio, los atacantes podrían intentar robar información como sus contraseñas, correos electrónicos o detalles de su tarjeta de crédito.

[Más información...](#)

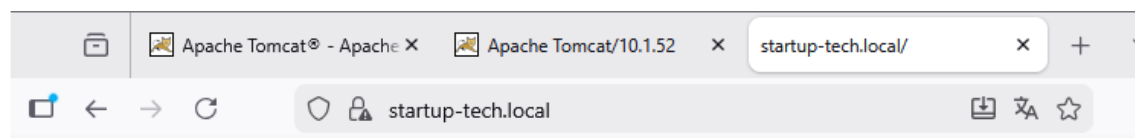
Retroceder (recomendado)

Avanzado...

Si aceptado pulsando el botón de avanzado podemos entrar y ver el certificado que ha cargado la página.



Comprobación de redirección de http a https



Bienvenido a Startup Tech - ENDIKA

Inspector	Consola	Depurador	Red	Editor de estilos	Rendimiento	Memoria	Almacenamiento
Filtrar las URL							
Todos HTML CSS JS XHR Tipografía Imágenes Medios WS Otros							
Estado	Método	Dominio	Archivo	Iniciador	Tipo	Transferido	
301	GET	startup-tech.io...	/	document	html	308 B	
200	GET	startup-tech.io...	/	document	html	cacheado	
200	GET	startup-tech.io...	favicon.ico	img	html	cacheado	

Ejercicio 3: Área privada y autenticación

Vamos a crear un directorio en `/var/www/startup-tech/privada` protegido para que solo las credenciales del usuario “empleado” y password “acceso” puedan acceder.

Lo primero es crear el fichero para almacenar las password y establecer los permisos de forma adecuada.

```
ubuntu26@ubuntuexamen: /etc/apache2/sites-available
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ sudo mkdir -p /etc/apache2/htpasswd
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ sudo htpasswd -c /etc/apache2/htpasswd/startup-tech.local empleado
New password:
Re-type new password:
Adding password for user empleado
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ sudo chown www-data: -R /etc/apache2/htpasswd && sudo chmod 400 /etc/apache2/htpasswd/startup-tech.local
```

Ahora creamos la carpeta privada

```
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ sudo mkdir -p /var/www/startup-tech/privado
```

Vamos a copiar el html que teníamos y cambiarle algo para que se note.

Vamos añadir las directivas para impedir que se pueda acceder sin contraseñas en el virtualhost para ese directorio.

```
ubuntu26@ubuntuexamen: /etc/apache2/sites-available
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ sudo nano /etc/apache2/^C
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ sudo nano startup-tech.local.conf
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ sudo systemctl restart apache2
ubuntu26@ubuntuexamen:/etc/apache2/sites-available$ cat startup-tech.local.conf
<VirtualHost *:80>
    ServerName startup-tech.local
    ServerAlias www.startup-tech.local

    ServerAdmin endika@localhost

    CustomLog ${APACHE_LOG_DIR}/startup_tech_local-access.log combined

    RewriteEngine On
    #Redireccion de trafico http a https
    RewriteCond %{HTTPS} off
    RewriteRule ^/(.*)$ https://%{SERVER_NAME}/$1 [R=301,L]
</VirtualHost>

<VirtualHost *:443>
    ServerName startup-tech.local
    ServerAlias www.startup-tech.local

    ServerAdmin endika@localhost

    DocumentRoot /var/www/startup-tech/publico

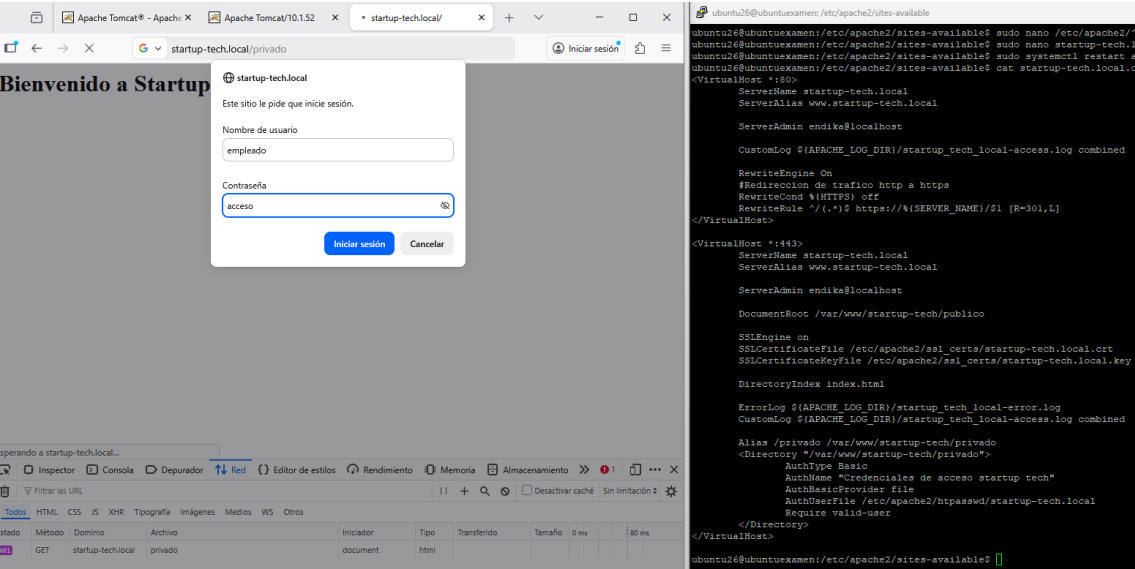
    SSLEngine on
    SSLCertificateFile /etc/apache2/ssl_certs/startup-tech.local.crt
    SSLCertificateKeyFile /etc/apache2/ssl_certs/startup-tech.local.key

    DirectoryIndex index.html

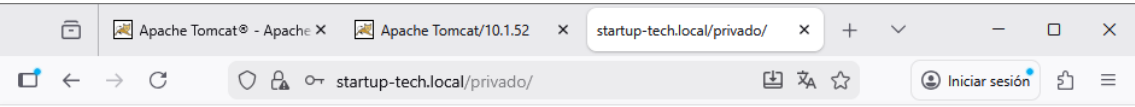
    ErrorLog ${APACHE_LOG_DIR}/startup_tech_local-error.log
    CustomLog ${APACHE_LOG_DIR}/startup_tech_local-access.log combined

    Alias /privado /var/www/startup-tech/privado
    <Directory "/var/www/startup-tech/privado">
        AuthType Basic
        AuthName "Credenciales de acceso startup tech"
        AuthBasicProvider file
        AuthUserFile /etc/apache2/htpasswd/startup-tech.local
        Require valid-user
    </Directory>
</VirtualHost>
```

Verificamos que funciona



Una vez introducida la password vemos que hemos conseguido entrar en la zona privada.



Bienvenido a la zona PRIVADA de Startup Tech - ENDIKA

Estado	Método	Dominio	Archivo	Iniciador	Tipo	Transferido	Tamaño	0 ms	1,37 min
301	GET	startup-tech.lo...	privado	document	html	335 B	63 B		182751 ms
200	GET	startup-tech.lo...	/privado/	document	html	346 B	63 B		5 ms
404	GET	startup-tech.lo...	favicon.ico	img	html	497 B	281 B		2 ms

Apartado C

Ejercicio1: Preparar entorno

El war nos lo ha proporcionado el examinado, pero hay que realizar ciertas configuraciones para que podamos usar el manager de tomcat.

Añadir usuario en /opt/tomcat/conf/tomcat-users.xml

```
ubuntu26@ubuntuexamen: /etc/apache2/sites-available
GNU nano 7.2 /opt/tomcat/conf/tomcat-users.xml
<?xml version="1.0" encoding="UTF-8"?>
<!--
Licensed to the Apache Software Foundation (ASF) under one or more
contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License. You may obtain a copy of the License at

http://www.apache.org/licenses/LICENSE-2.0

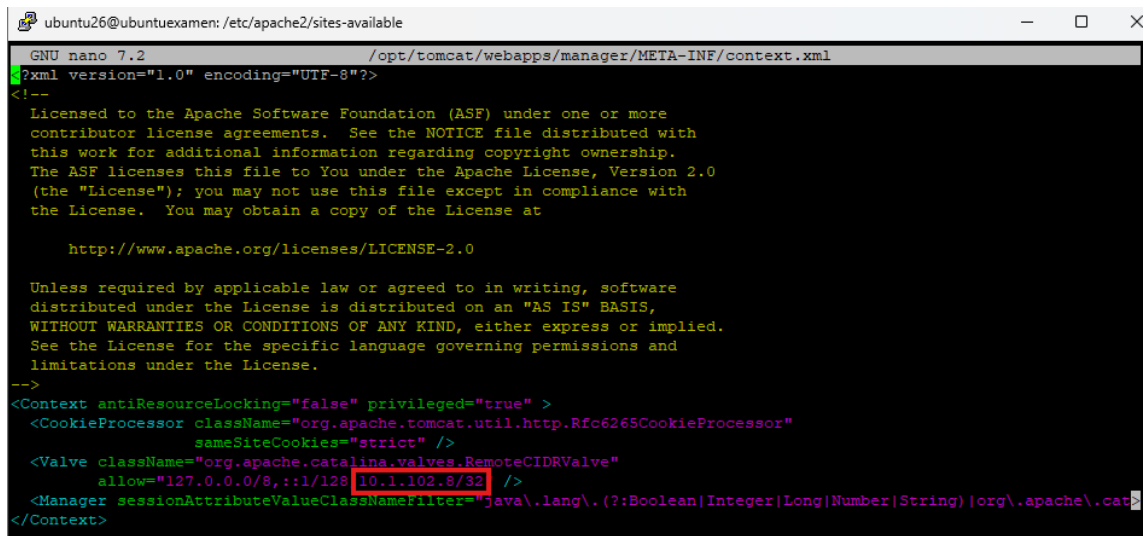
Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<tomcat-users xmlns="http://tomcat.apache.org/xml"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://tomcat.apache.org/xml tomcat-users.xsd"
version="1.0">
<!--
By default, no user is included in the "manager-gui" role required
to operate the "/manager/html" web application. If you wish to use this app,
you must define such a user - the username and password are arbitrary.

Built-in Tomcat manager roles:
- manager-gui - allows access to the HTML GUI and the status pages
- manager-script - allows access to the HTTP API and the status pages
- manager-jmx - allows access to the JMX proxy and the status pages
- manager-status - allows access to the status pages only

The users below are wrapped in a comment and are therefore ignored. If you
wish to configure one or more of these users for use with the manager web
application, do not forget to remove the <!-- .. --> that surrounds them. You
will also need to set the passwords to something appropriate.
-->
<!--
<user username="admin" password="<must-be-changed>" roles="manager-gui"/>
<user username="robot" password="<must-be-changed>" roles="manager-script"/>
-->
<!--
The sample user and role entries below are intended for use with the
examples web application. They are wrapped in a comment and thus are ignored
when reading this file. If you wish to configure these users for use with the
examples web application, do not forget to remove the <!-- .. --> that surrounds
them. You will also need to set the passwords to something appropriate.
-->
<!--
<role rolename="tomcat"/>
<role rolename="role1"/>
<user username="tomcat" password="<must-be-changed>" roles="tomcat"/>
<user username="both" password="<must-be-changed>" roles="tomcat,role1"/>
<user username="role1" password="<must-be-changed>" roles="role1"/>
-->
<role rolename="manager-gui" />
<user username="endika" password="examen26" roles="manager-gui" />
</tomcat-users>
```

[Wrote 58 lines]

Adicionalmente como no accedemos como localhost debemos incluir la IP del equipo o el rango de IP's desde el que nos conectamos.

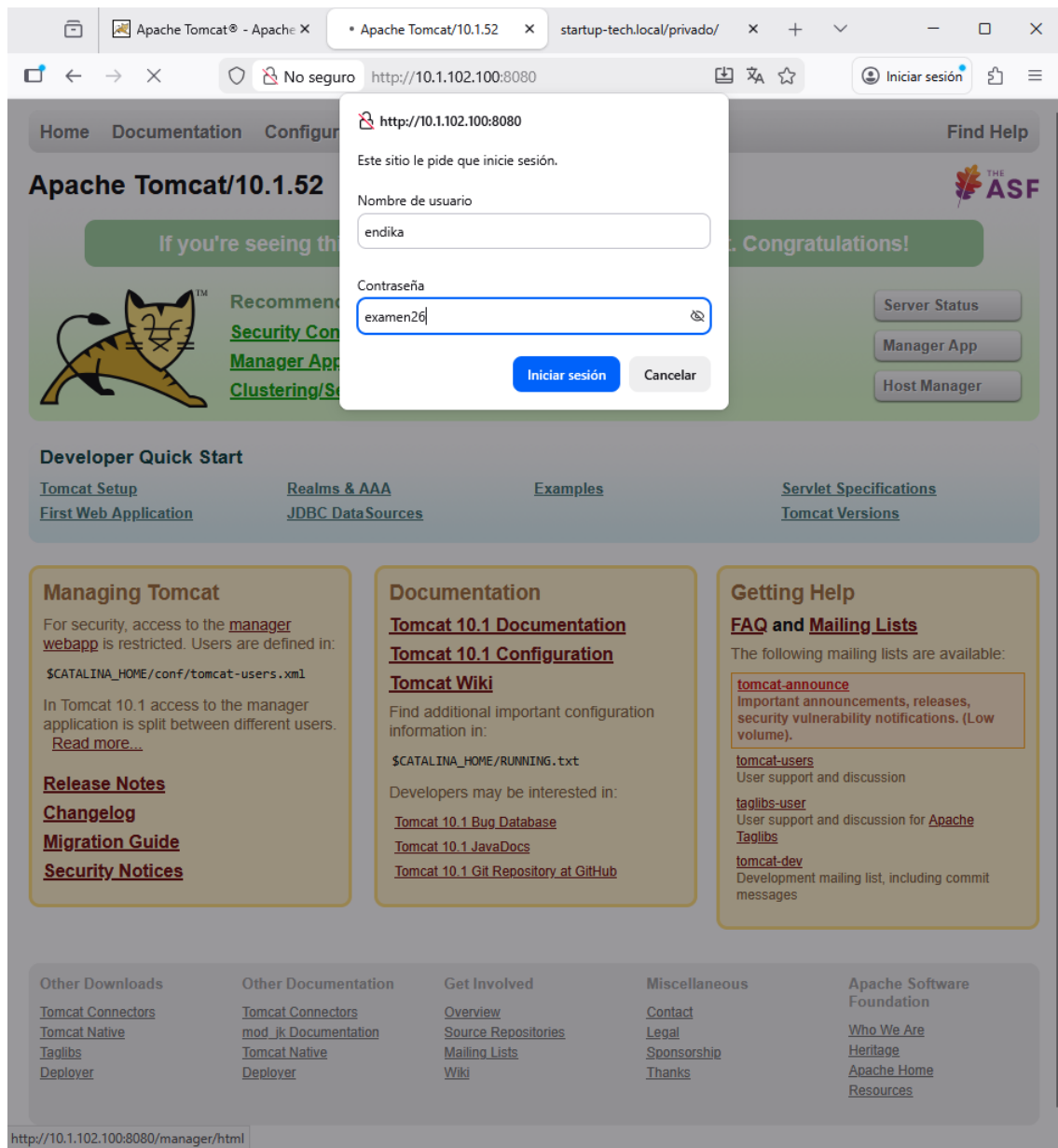


```
ubuntu26@ubuntuexamen: /etc/apache2/sites-available
GNU nano 7.2 /opt/tomcat/webapps/manager/META-INF/context.xml
?xml version="1.0" encoding="UTF-8"?>
<!--
Licensed to the Apache Software Foundation (ASF) under one or more
contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License. You may obtain a copy of the License at

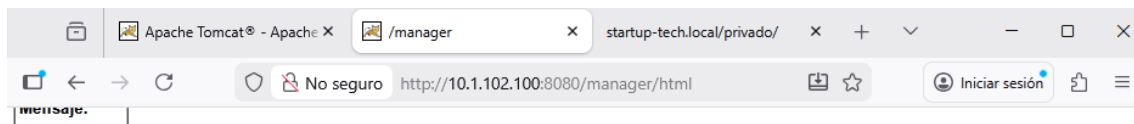
    http://www.apache.org/licenses/LICENSE-2.0

Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
    sameSiteCookies="strict" />
  <Valve className="org.apache.catalina.valves.RemoteCIDRValve"
    allow="127.0.0.0/8,::1/128,10.1.102.8/32" />
  <Manager sessionAttributeValueClassNameFilter="java\.lang\.(?:Boolean|Integer|Long|Number|string)|org\.apache\.cat
</Context>
```

Comprobación de que podemos desplegar la aplicación.



Desplegamos el war



Gestor			
Listar Aplicaciones	Ayuda HTML de Gestor	Ayuda de Gestor	Estado de Servidor

Aplicaciones					
Ruta	Versión	Nombre a Mostrar	Ejecutándose	Sesiones	Comandos
/	Ninguno especificado	Welcome to Tomcat	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/docs	Ninguno especificado	Tomcat Documentation	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/examples	Ninguno especificado	Servlet and JSP Examples	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/host-manager	Ninguno especificado	Tomcat Host Manager Application	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/manager	Ninguno especificado	Tomcat Manager Application	true	1	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos

Desplegar	
Desplegar directorio o archivo WAR localizado en servidor	
Trayectoria de Contexto (opcional):	
Version (for parallel deployment):	
URL de archivo de Configuración XML:	
URL de WAR o Directorio:	
Desplegar	
Archivo WAR a desplegar	
Seleccione archivo WAR a cargar Examinar... hola-mundo.war	
Desplegar	

Vemos que se ha desplegado

Mensaje:	OK
----------	----

Gestor			
Listar Aplicaciones	Ayuda HTML de Gestor	Ayuda de Gestor	Estado de Servidor

Aplicaciones					
Ruta	Versión	Nombre a Mostrar	Ejecutándose	Sesiones	Comandos
/	Ninguno especificado	Welcome to Tomcat	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/docs	Ninguno especificado	Tomcat Documentation	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/examples	Ninguno especificado	Servlet and JSP Examples	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/hola-mundo	Ninguno especificado		true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/host-manager	Ninguno especificado	Tomcat Host Manager Application	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/manager	Ninguno especificado	Tomcat Manager Application	true	1	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos

Parece que el .war no contiene archivos indexables.

